

$$\begin{aligned}
&= \varphi_0 \left(\frac{\xi_{\mathbf{L}_k}}{2} \right) \frac{1}{P} \sum_{j=0}^{P-1} \left[\sum_{l=-M}^M \varphi_0 \left(\frac{\xi_{\mathbf{L}_l}}{2} \right) \hat{u}_l^n e^{i\kappa_l x_{r_j}} \right] e^{-i\kappa(p_j+r_j)\Delta x} \\
&\neq \varphi_0(\xi_{\mathbf{L}_k}) \frac{1}{P} \sum_{j=0}^{P-1} u(t_n, x_{r_j}) e^{-i\kappa(p_j+r_j)\Delta x}
\end{aligned} \tag{6.19}$$

where $\kappa_l := 2\pi l/L$. Note that the dependency of $\xi_{\mathbf{L}_l}$ on l prevents us from factoring out the exponential term from the inner summation, such that the inverse Fourier transform into the brackets does not yield $u(t_n, x_{r_j})$. As a consequence, different stability functions are provided by the two variants of the semi-Lagrangian exponential scheme.

Therefore, by explicitly writing the spectral transforms involved in the integration of the semi-Lagrangian exponential schemes, it becomes clear why different discretizations of the linear term lead to different stability functions. We remark, however, that these differences are not necessarily linked to the fact that the equations are solved using a spectral method, but rather to the lack of commutation between the matrix exponential and the interpolation. For illustration, consider the equation

$$\frac{d\mathbf{u}}{dt} = \lambda_{\mathbf{L}} \mathbf{u} \tag{6.20}$$

defined on the (periodic) physical space, with $\mathbf{u} = (u_1, \dots, u_P)^T \in \mathbb{R}^P$ and $\lambda_{\mathbf{L}} = \text{diag}(\lambda_1, \dots, \lambda_P) \in \mathbb{R}^{P \times P}$ a diagonal matrix with at least one element λ_i different from the others. Consider also that the interpolation procedure consists simply of a one grid point shift to the right. Therefore, the integration of one time step of (6.20) using the two variants of the semi-Lagrangian exponential scheme yields respectively

$$\mathbf{u}^{n+1} = \varphi_0(\Delta t \lambda_{\mathbf{L}}) [\mathbf{u}^n]_* = \begin{pmatrix} \varphi_0(\Delta t \lambda_1) & & 0 \\ & \ddots & \\ 0 & & \varphi_0(\Delta t \lambda_P) \end{pmatrix} \begin{pmatrix} u_P^n \\ u_1^n \\ \vdots \\ u_{P-1}^n \end{pmatrix} = \begin{pmatrix} \varphi_0(\Delta t \lambda_1) u_P^n \\ \varphi_0(\Delta t \lambda_2) u_1^n \\ \vdots \\ \varphi_0(\Delta t \lambda_P) u_{P-1}^n \end{pmatrix} \tag{6.21}$$

and

$$\begin{aligned}
\mathbf{u}^{n+1} &= \varphi_0 \left(\frac{\Delta t \lambda_{\mathbf{L}}}{2} \right) \left[\varphi_0 \left(\frac{\Delta t \lambda_{\mathbf{L}}}{2} \right) \mathbf{u}^n \right]_* \\
&= \begin{pmatrix} \varphi_0(\Delta t \lambda_1/2) & & 0 \\ & \ddots & \\ 0 & & \varphi_0(\Delta t \lambda_P/2) \end{pmatrix} \left[\begin{pmatrix} \varphi_0(\Delta t \lambda_1/2) u_1^n \\ \varphi_0(\Delta t \lambda_2/2) u_2^n \\ \vdots \\ \varphi_0(\Delta t \lambda_P/2) u_P^n \end{pmatrix} \right]_* \\
&= \begin{pmatrix} \varphi_0(\Delta t \lambda_1/2) & & 0 \\ & \ddots & \\ 0 & & \varphi_0(\Delta t \lambda_P/2) \end{pmatrix} \left[\begin{pmatrix} \varphi_0(\Delta t \lambda_P/2) u_P^n \\ \varphi_0(\Delta t \lambda_1/2) u_1^n \\ \vdots \\ \varphi_0(\Delta t \lambda_{P-1}/2) u_{P-1}^n \end{pmatrix} \right]_*
\end{aligned}$$

$$= \begin{pmatrix} \varphi_0(\Delta t \lambda_1/2) \varphi_0(\Delta t \lambda_P/2) u_P^n \\ \varphi_0(\Delta t \lambda_2/2) \varphi_0(\Delta t \lambda_1/2) u_1^n \\ \vdots \\ \varphi_0(\Delta t \lambda_P/2) \varphi_0(\Delta t \lambda_{P-1}/2) u_{P-1}^n \end{pmatrix} \quad (6.22)$$

with the two schemes being equivalent only if $\lambda_{\mathbf{L}}$ is a multiple of the identity matrix.

Note that, except for the case in which the velocity field is constant, determining the stability regions of SE** analytically, using (6.14) and (6.19), is not feasible. In particular, we are not able to obtain an expression in the form $\hat{u}_k^{n+1} = A(\xi_{\mathbf{L}}, \xi_{\mathbf{N}}) \hat{u}_k^n$ for each k separately, and $\hat{u}_k^{n+1}$ depends *a priori* on all wavenumber modes. It means that we are no longer in the framework of a linear stability analysis, even if the problem (6.3) itself is linear, with this *nonlinearity* being a consequence of the lack of commutation between the exponential and interpolation operators. We choose therefore to assess the stability properties of the schemes through numerical simulations, as presented in Section 7.3.

7. NUMERICAL TESTS

In this section, we perform a set of numerical simulations of the SWE on the rotating sphere to evaluate the proposed modifications of the semi-Lagrangian exponential methods and to compare them with their original versions and with other relevant numerical schemes in this study, namely SL-SI-SETTLS and the Eulerian exponential integration methods. We begin by verifying the numerical orders of convergence using benchmark test cases with increasing complexity; then, we conduct an empirical stability study; finally, we consider a relatively challenging test to evaluate the methods qualitatively and in terms of computational cost, including a study of the application of hyperviscosity approaches.

7.1. Numerical verification of the convergence order considering the advection equation

The goal of this section is to provide an initial numerical validation of the truncation error analyses developed in Sections 4 and 5.1, considering the governing equations in the Lagrangian framework and containing only the linear term on the right-hand side.

We consider first the scalar, one-dimensional equation (4.2), in the periodic domain $\Omega = [0, 10]$, integrated from $t = 0$ to $t = 10$, and a Gaussian curve centered at $x = 5$ as initial solution. We consider time step sizes ranging between $\Delta t = 2^{-5}$ and $\Delta t = 1$, and the spatial domain is discretized considering $P + 1 = 2049$ equally spaced points. We consider the first- and second-order schemes SE12 and SE22 (without the nonlinear terms), and, for comparison, we also integrate using SL-SI-SETTLS. A reference solution is computed using a 4th-order explicit Runge Kutta method with time step size $\Delta t = 2^{-5}/10$ and the same spatial resolution $P + 1 = 2049$, with the spatial derivatives being computed using the Fourier spectral method with spectral resolution $M = 1024$. Figure 4a presents the relative L_2 errors in two cases, respectively with constant $L(x) \equiv 1$, and variable $L(x) = \sin(x)$. In the former, both semi-Lagrangian exponential methods compute the exact solution (up to the error due to the spectral resolution in the reference simulation), since the integration factor (3.14) is exactly $P_n(t_{n+1}) = \varphi_0(\Delta t L)$. In the latter case, in which L depends on space (thus on time, along each Lagrangian trajectory), the first- and second-order accuracies derived analytically are observed.

In Figure 4b, we present the same results in the vectorial case (5.4), with $N = 2$ and one spatial dimension. The same discretization parameters as in the scalar examples are considered. We perform simulations for two choices of $\mathbf{L}(x)$, respectively

$$\mathbf{L}(x) = \mathbf{L}_1(x) := \begin{pmatrix} \sin(x) & \cos(x) \\ \cos(x) & \sin(x) \end{pmatrix}, \quad \mathbf{L}(x) = \mathbf{L}_2(x) := \begin{pmatrix} \sin(x) & \sin(x) \\ \sin(x) & \cos(x) \end{pmatrix} \quad (7.1)$$

with the property that $\mathbf{L}_1(x)$ and $\mathbf{L}_1(y)$ commutes for every x, y , which does not hold for $\mathbf{L}_2$. In both cases, SE12 provides first-order accuracy, due to the approximation of the integrals considering the space-dependent linear operators to be constant along the trajectories. On the other hand, SE22 is second-order accurate in both cases, including the one in which the linear operator does not commute along the trajectory. The simulations presented in the following paragraphs, considering the SWE on the rotating sphere, for which the commutation assumption does not hold, confirm these results.

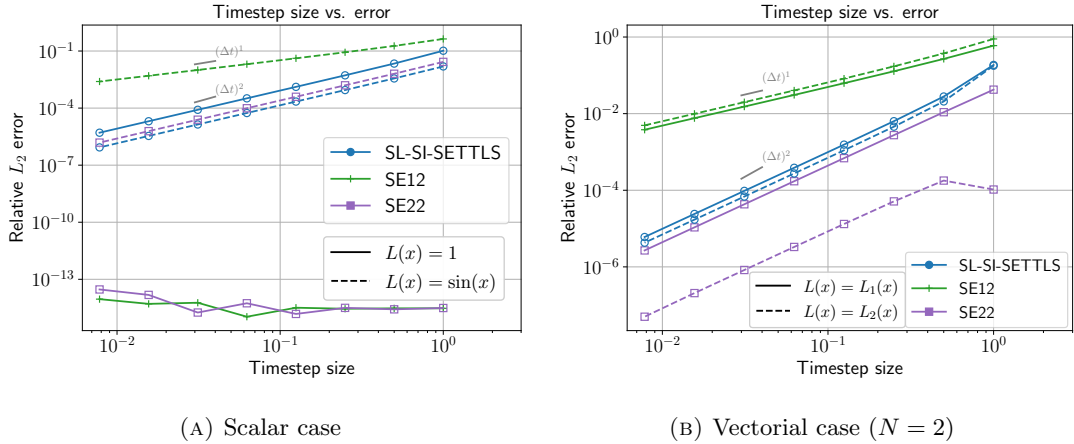


FIGURE 4. Numerical verification of the convergence order of the semi-Lagrangian schemes considered in this work (relative L_2 error *vs* time step size), considering governing equations with only advective and linear terms.

7.2. Numerical verification of the convergence order considering the SWE on the rotating sphere

We consider here three test cases to verify the orders of convergence determined analytically. To provide a more detailed insight into the stability properties of the time integration schemes, we present, for each simulation, the CFL numbers related to gravity wave propagation and advection. We consider the characteristic speed $c = \bar{\Phi}^{1/2} + |\bar{u}|$, where the square root of the mean geopotential, $\bar{\Phi}^{1/2} = \sqrt{gh}$, defines the mean external gravity wave speed, and $\bar{u}$ is the maximum advective velocity magnitude defined by the initial conditions. A reference CFL number is then defined as $c\Delta t/\Delta x$, where Δx is the longitudinal grid size along the equator.

The test cases considered here are: